

UNCERTAINTY MEETS INCONSISTENCY

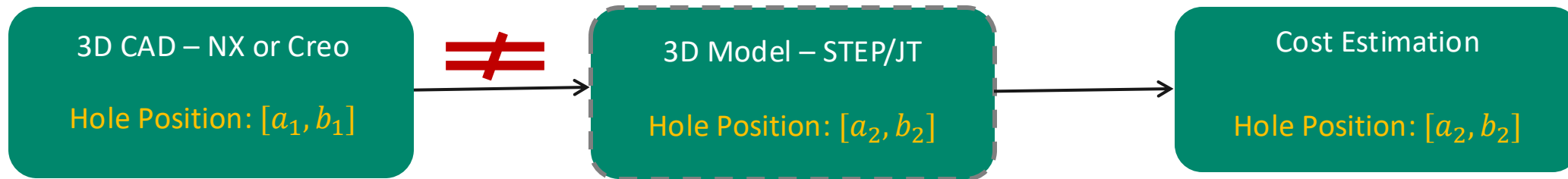


INCONSISTENCY 1: MANUAL COST ESTIMATION FROM CAD



- The STEP/JT model becomes outdated because exports are done manually.
- Cost estimation tools rely on this outdated representation.
- Recent CAD modifications are not reflected in the neutral model.
- Geometric information used for costing may deviate from the actual design.

UNCERTAINTY 1: GEOMETRY MISMATCH BETWEEN CAD AND STEP/JT



Two models represent inconsistent uncertainties

- CAD contains uncertain hole position $[a_1, b_1]$
- STEP/JT contains a different interval after manual update $[a_2, b_2]$

Inconsistency may increase, reduce, or shift uncertainty

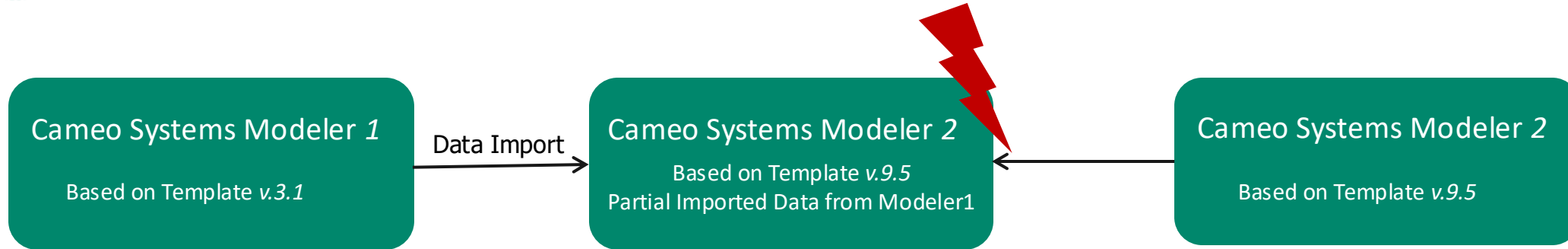
- Increase: $[a_2, b_2] > [a_1, b_1]$
- Reduced: $[a_2, b_2] < [a_1, b_1]$
- Shift: $[a_2, b_2] \gg [a_1, b_1]$ or $[a_2, b_2] \ll [a_1, b_1]$

Uncertainty Representation

Location: Model Outcome
Source: Human in the Loop
Kind: Measurement
Effect: continuous non-deterministic

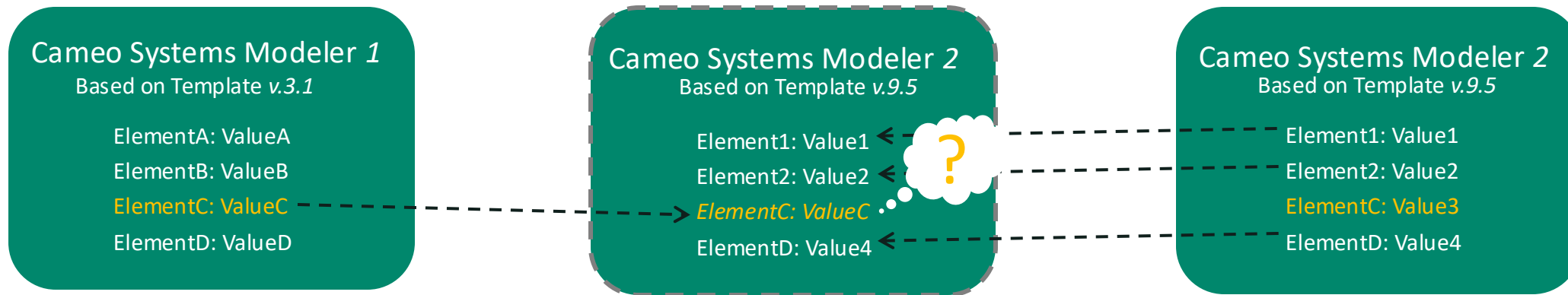
INCONSISTENCY 2:

TOOL ISSUES IN THE SAME MBSE MODEL



- Two MBSE models were created using incompatible template versions.
- Merging results in invalid or unreadable model structures.
- Imported elements no longer match the expected stereotype definitions.
- The resulting integrated model becomes structurally inconsistent.

UNCERTAINTY 2: SEMANTIC INTERPRETATION UNCERTAINTY



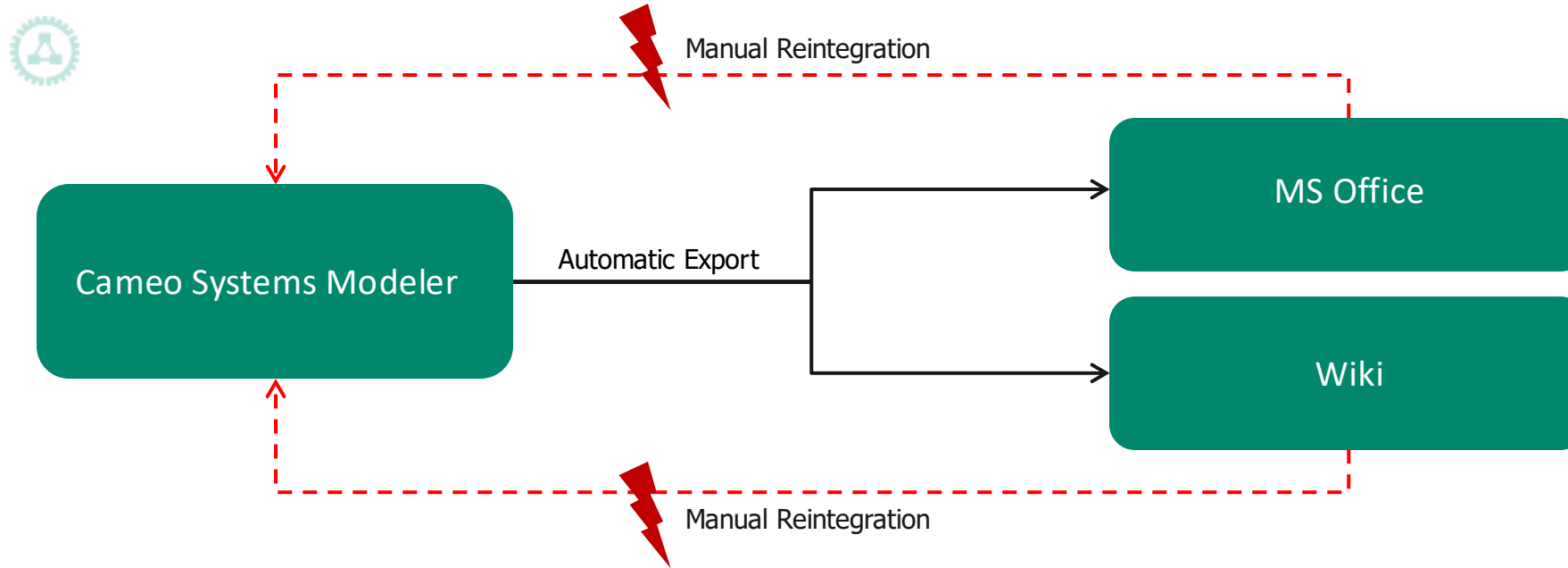
- Template v3.1 and v9.5 define different semantics to ElementC
- Import preserves ValueC, but its meaning changes in v.9.5
- The modeler cannot determine which semantic definition is applied

Uncertainty Representation

Location: Model Structure
Source: Model
Kind: Belief
Effect: discrete non-deterministic

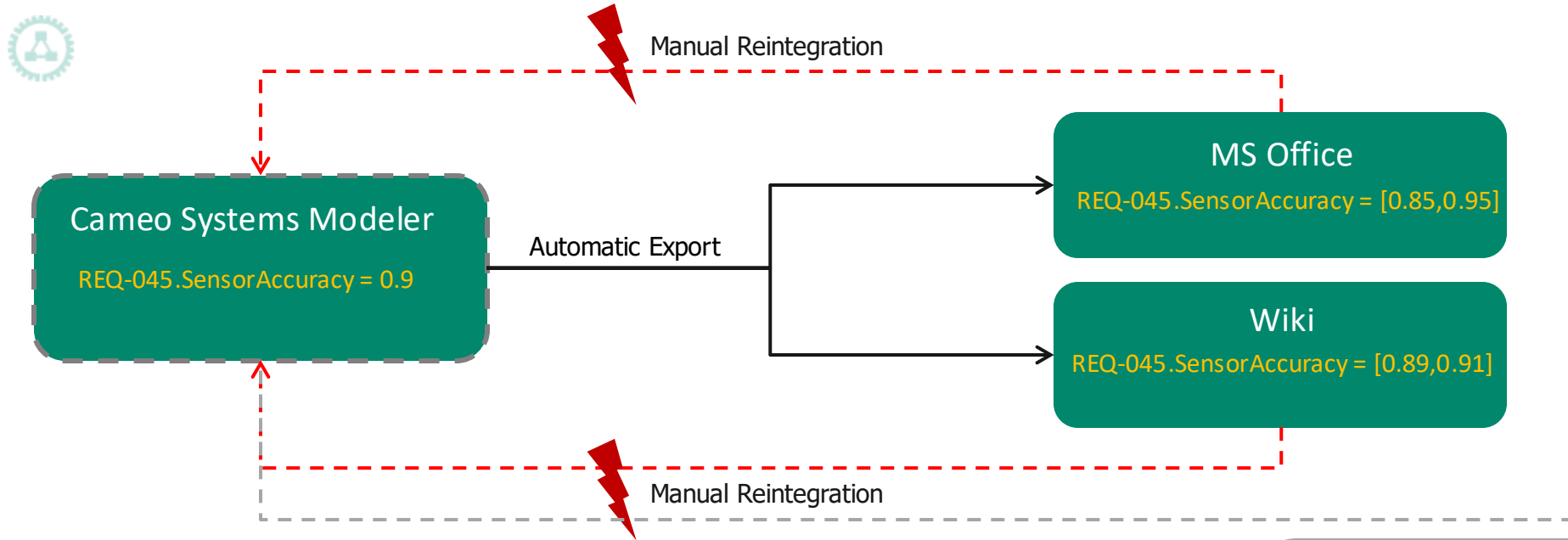
INCONSISTENCY 3:

LINKING OF MBSE MODELS AND PLAIN TEXT



- Cameo content is automatic exported to both MS Office and Wiki models for discussions
- Stakeholders update the Office and Wiki models independently during discussions.
- No automated or defined process to transfer changes back into Cameo -> manual reintegration
- Cameo, Office, and Wiki models drift apart over time, leading to inconsistent system knowledge.

UNCERTAINTY 3: MANUAL INFORMATION FLOW UNCERTAINTY



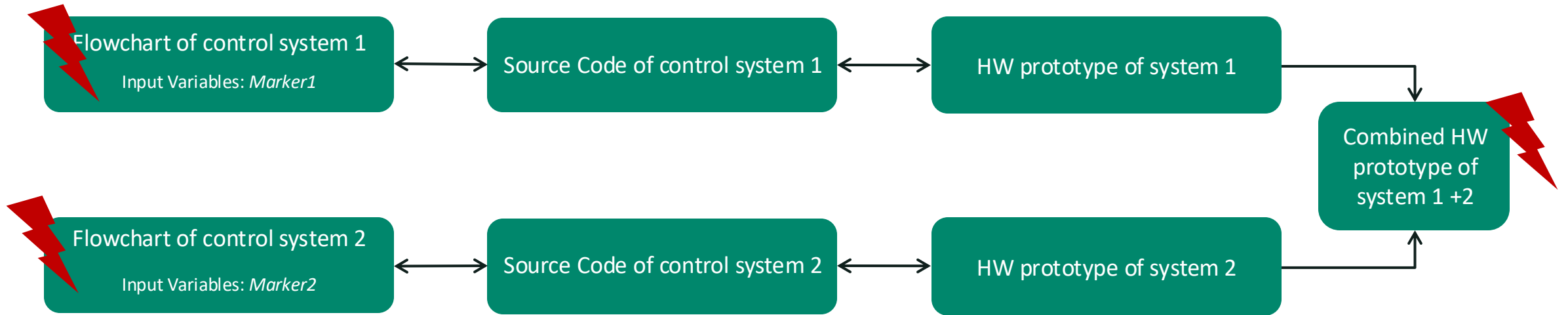
- Cameo specifies REQ-045.SensorAccuracy = 0.9,
- MS Office and Wiki use intervals update [0.85,0.95] and [0.89,0.91].
- Unclear which requirement formulation is authoritative.

Uncertainty Representation

Location: Model Structure
Source: Human in the Loop
Kind: Belief
Effect: discrete non-deterministic

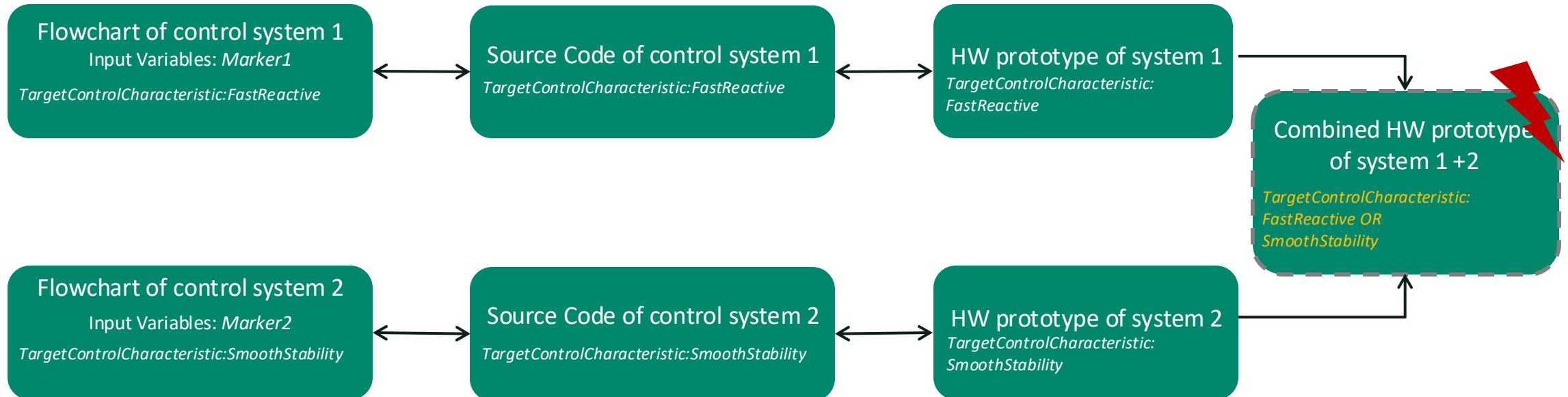
INCONSISTENCY 4:

INTEGRATION CONFLICTS IN EARLY DEVELOPMENT MODELS



- Controllers are synchronized but follow different principal solutions.
- Their assumptions and internal markers interact in unforeseen ways.
- The combined behaviour deviates from what each individual model predicts.
- Emergent effects appear that were not captured in separate designs.

UNCERTAINTY 4: CONFLICTING TARGET CONTROL CHARACTERISTICS



- It is uncertain which control characteristic the integrated system should follow.
- The controllers impose incompatible behavioural assumptions.
- The effective control behaviour becomes indeterminate during operation
- This unresolved uncertainty contributes to unstable system behaviour

Uncertainty Representation

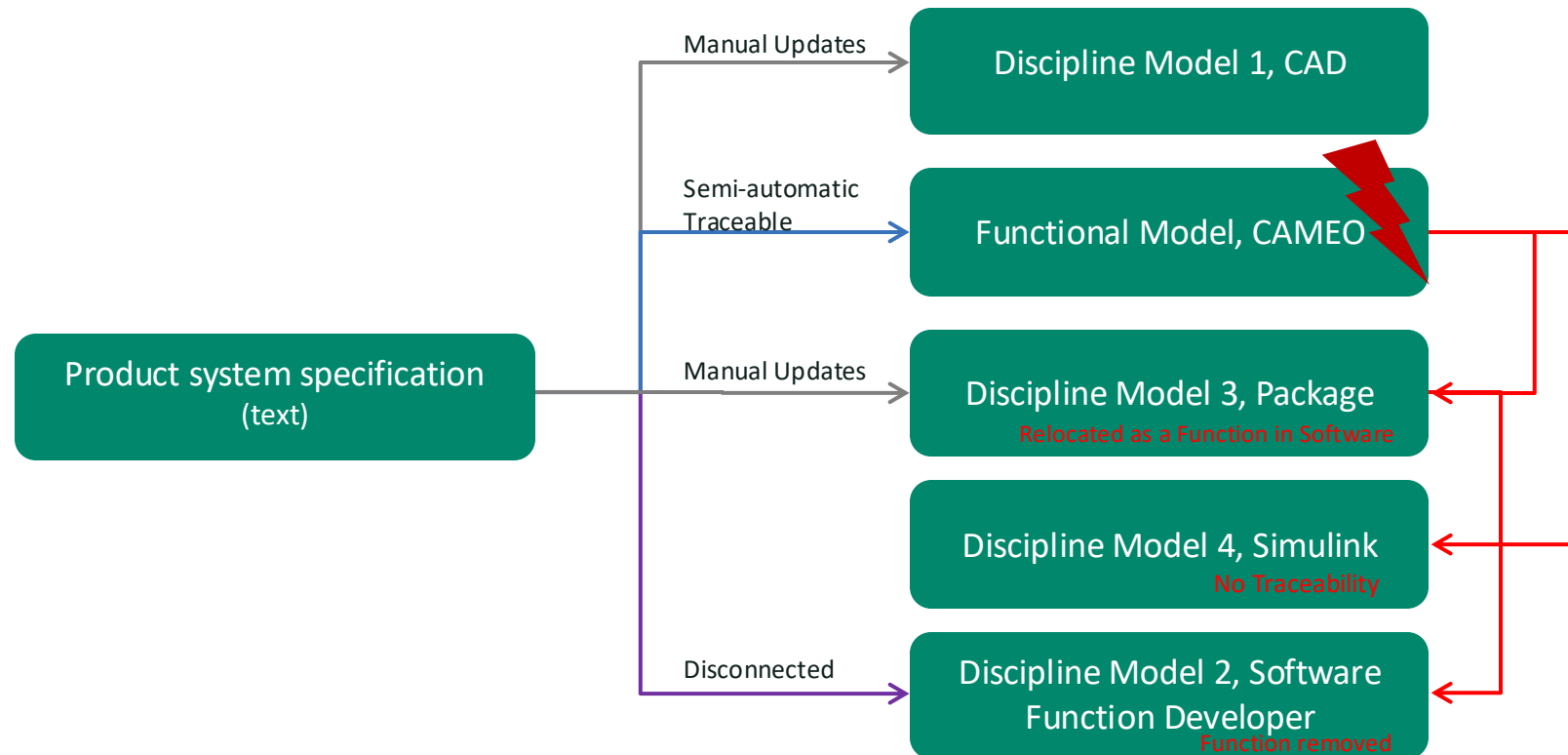
Location: Decision Making

Source: Goals

Kind: Behaviour

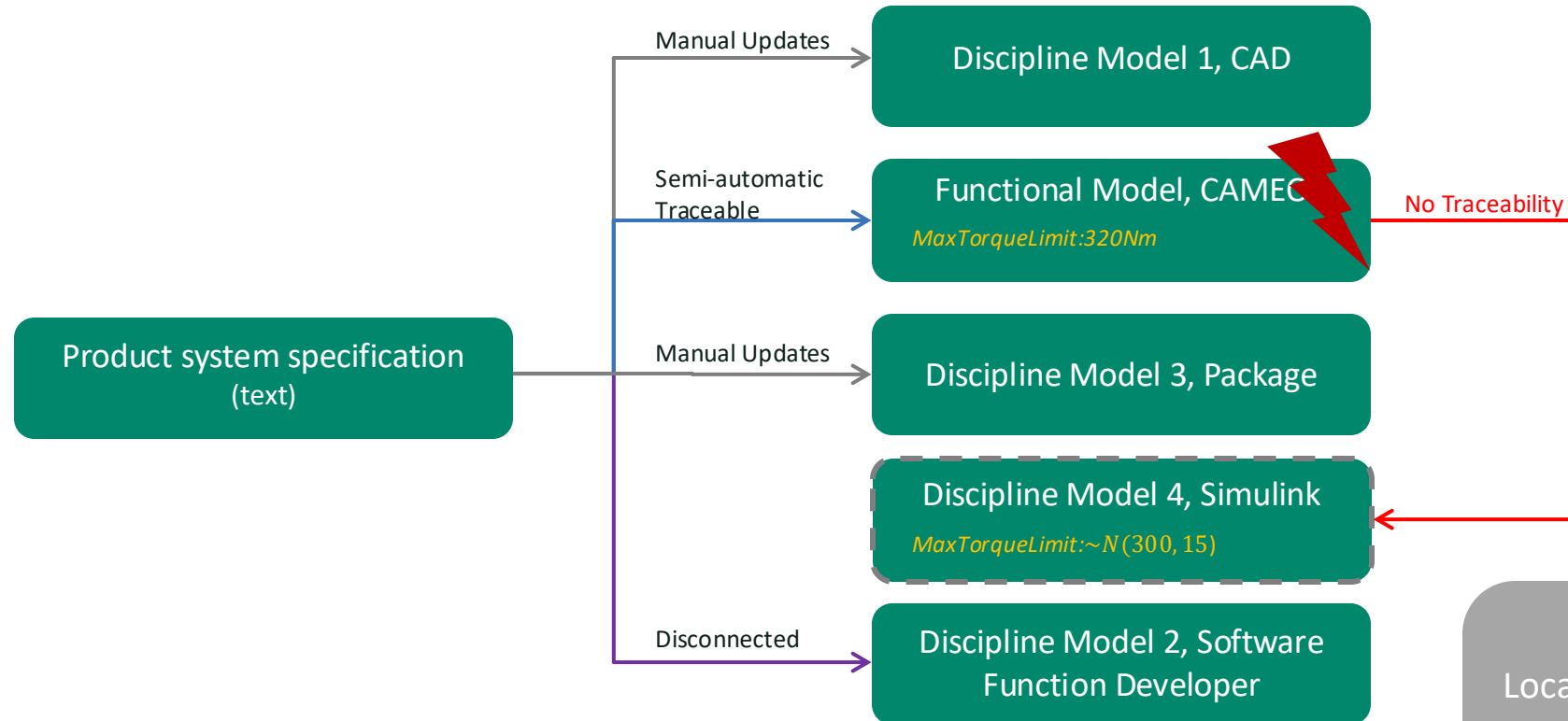
Effect: discrete non-deterministic

INCONSISTENCY 5: NO AUTOMATIC FEEDBACK INTO REQUIREMENT MODELS



- Architectural updates propagate inconsistently across domains, leading to diverging model states
- Missing traceability results in unintended function shifts and uncoordinated removals.
- Functions shift or disappear across models because teams work with outdated or disconnected architectural information.

UNCERTAINTY 5: AMBIGUOUS TORQUE LIMIT



Uncertainty Representation

Location:

Source:

Kind:

Effect: continuous probabilistic

- Missing traceability leaves the correct torque limit unknown in the Simulink model.
- MaxTorqueLimit becomes uncertain and is represented as a probability distribution.
- The uncertainty arises only because the architectural value cannot be retrieved.

INCONSISTENCY 6:

INCONSISTENCY IN VARIANT CONFIGURATION FOR CUSTOMER



- TR 3.1 forbids combining Adaptive Cruise Control (ACC) and Lane Keeping Assist (LKA) for this vehicle variant.
- CR 4.2 still offers a Driver Assistance Package including ACC + LKA.
- It becomes unclear whether the technical restriction or the commercial offering should apply.
- Missing consistency checks and outdated tools allow this contradiction to persist.

UNCERTAINTY 6:

SENSING PERFORMANCE UNCERTAINTY



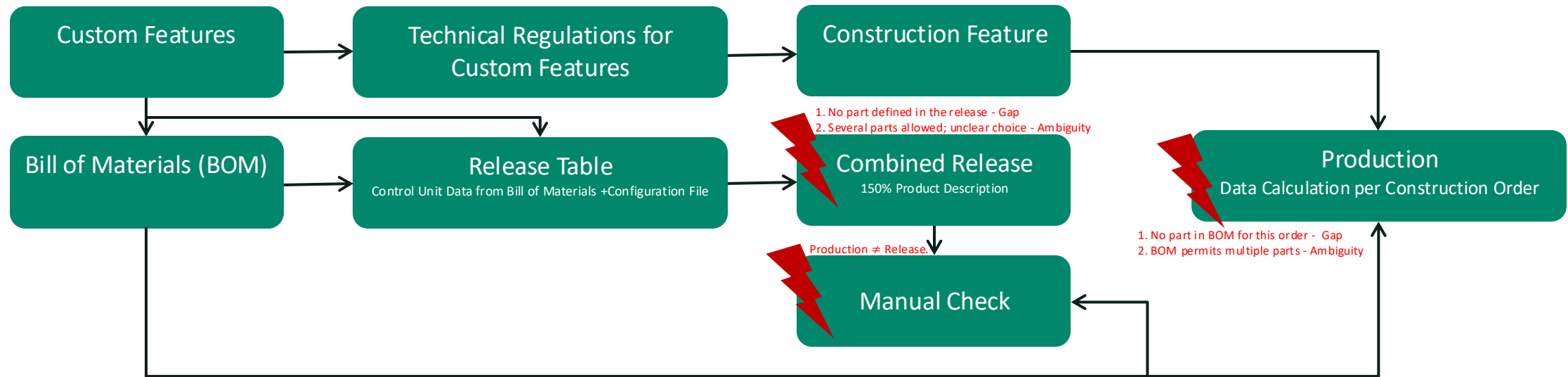
- ACC sensor range varies between 95–118 m due to environmental and calibration factors.
- TR prohibits ACC + LKA because safe operation requires a range > 105 m.
- Due to the inconsistency, CR ignores this and still offers the combined package.

Uncertainty Representation

Location: Parameter
Source: Environment
Kind: Measurement
Effect: continuous non-deterministic

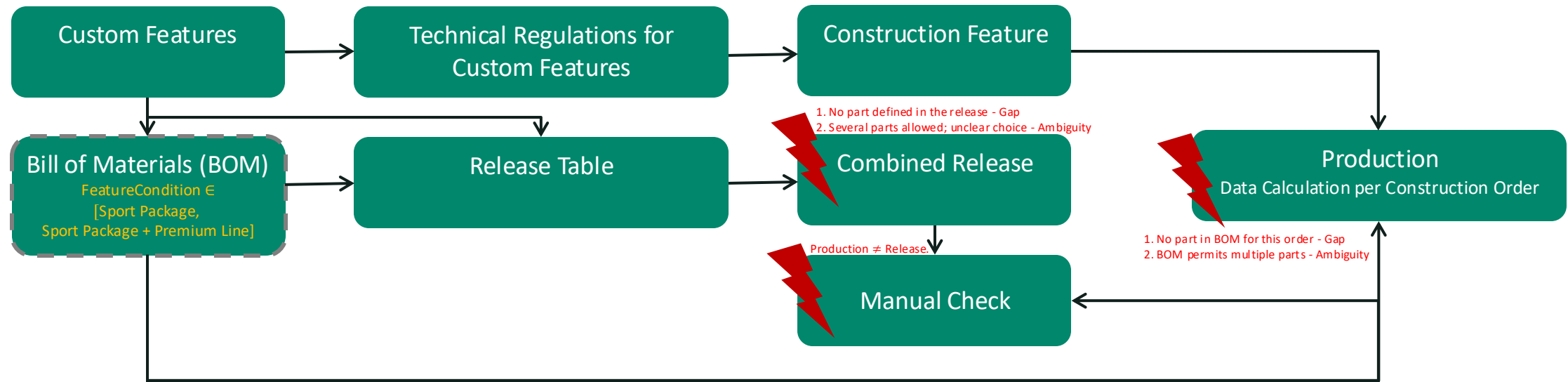
INCONSISTENCY 7:

INCONSISTENCY IN VARIANT CONFIGURATION FOR PRODUCTION



- Several models define which component must be installed for a given configuration.
- Misaligned rules in Combined Release and BOM cause gaps or ambiguities in part selection.
- Production cannot determine a unique or valid component for the control unit.
- Manual checks reveal that the built product does not match the frozen release.

UNCERTAINTY 7: CONFLICTING REGULATION BELIEF UNCERTAINTY



- The correct feature condition in the BOM is unclear ("Sport Package" vs. "Sport + Premium").
- Misaligned artifacts make it unclear which condition is correct.
- Uncertainty causes the construction order to match either no BOM line (gap) or several (ambiguity).
- Production may choose the wrong part, leading to a mismatch in the final vehicle.

Uncertainty Representation

Location: Model Input

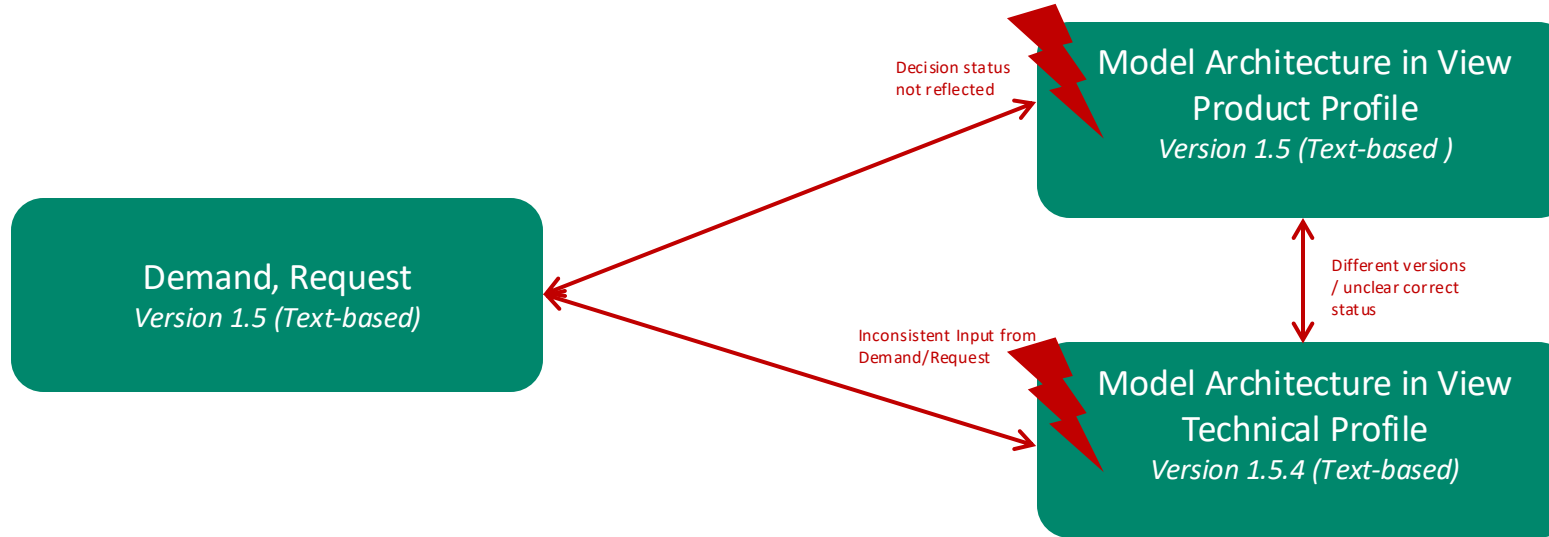
Source: Goals

Kind: Belief

Effect: discrete non-deterministic

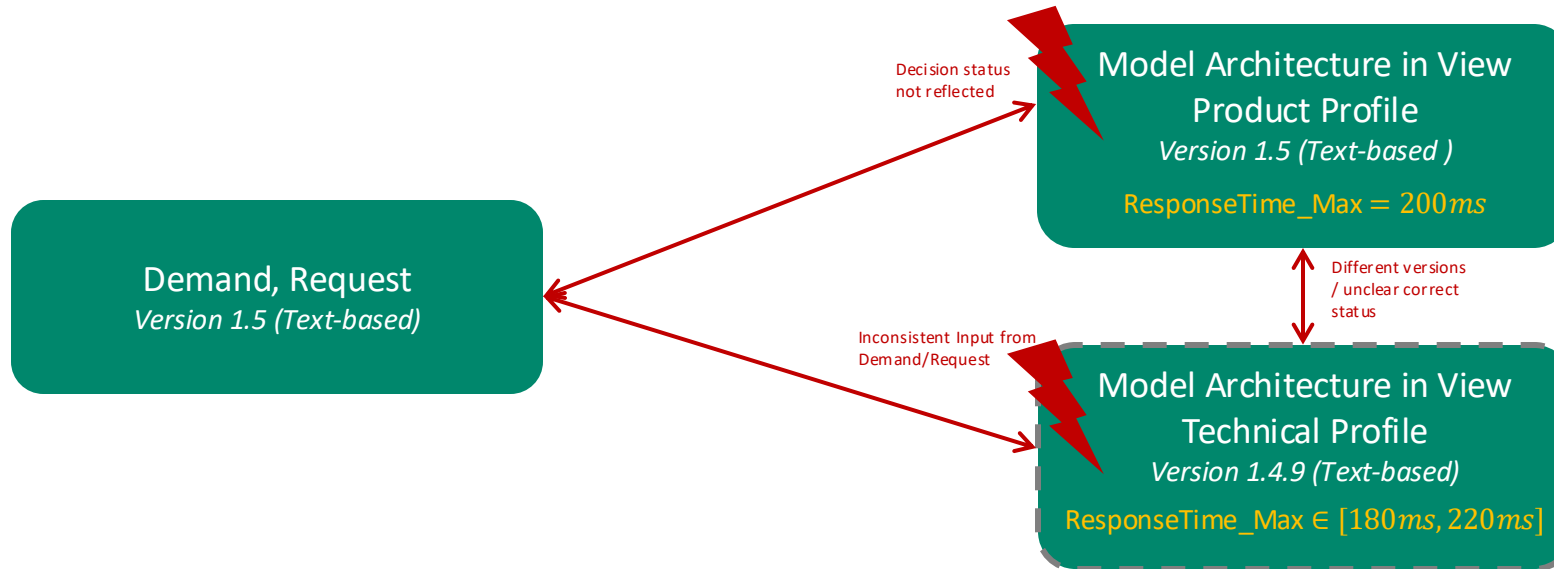
INCONSISTENCY 8:

INCONSISTENCY BETWEEN NEED REPRESENTATIONS & REQUIREMENTS



- The decision status from the Demand/Request is not consistently reflected in the Product Profile documentation.
- The specification is forwarded inconsistently, causing misalignment when deriving the Technical Profile.
- The Product and Technical Profiles contain different document versions, making it unclear which status is correct.
- These divergences lead to discussions among engineers about the correct architecture status and slow down coordination.

UNCERTAINTY 8: CONFLICTING REGULATION BELIEF UNCERTAINTY



- The Product Profile already specifies a fixed requirement: ResponseTime_Max = 200 ms.
- The Technical Profile still uses an uncertain performance range of [180 ms, 220 ms].
- The profiles therefore document different values for the same parameter.
- This difference results in an inconsistency between the two architecture views.

Uncertainty Representation

Location: Model Input

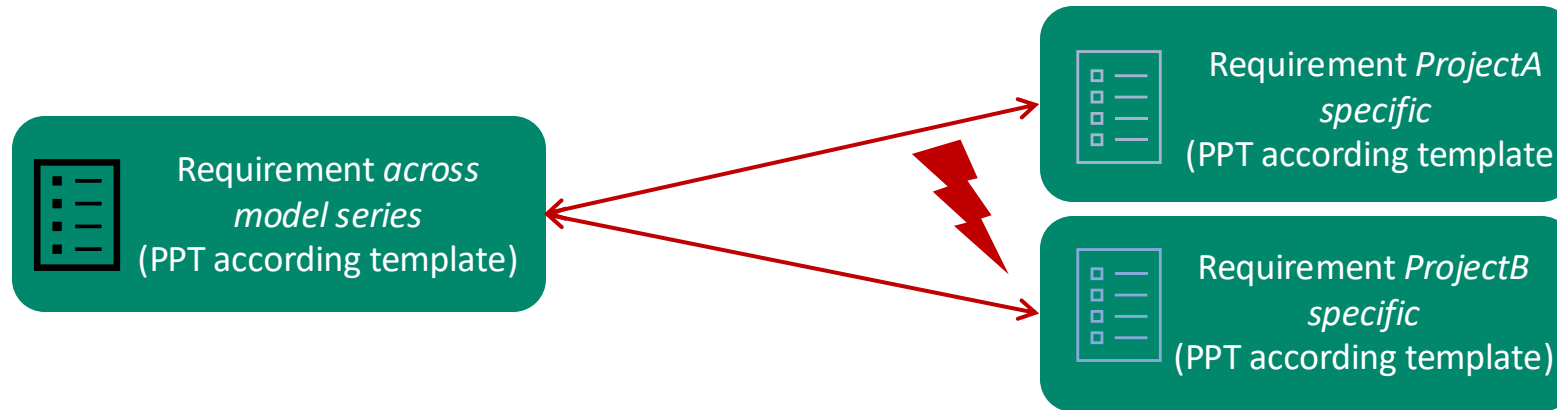
Source: Goals

Kind: Belief

Effect: continuous non-deterministic

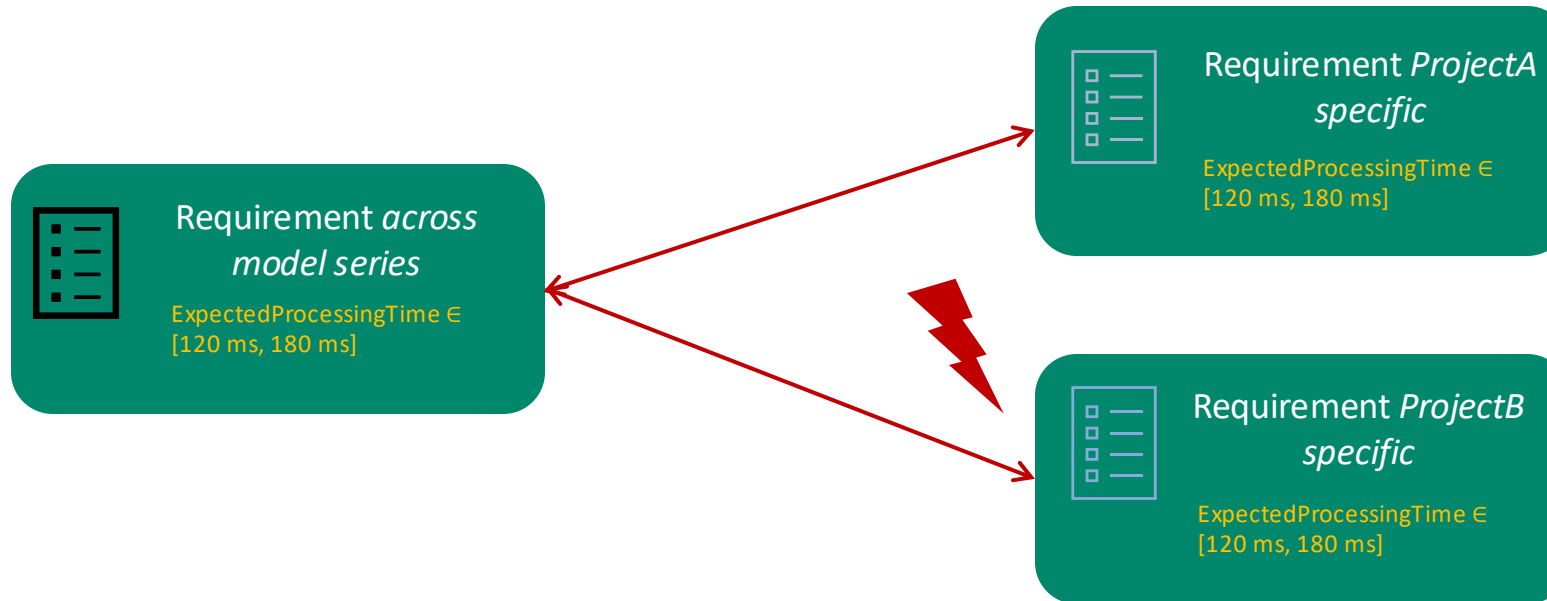
INCONSISTENCY 9:

INCONSISTENCY BETWEEN MODEL SERIES AND SUB-PROJECTS



- The stakeholder-level requirement template defines the intended structure for use across the model series
- Individual projects adapt this template locally, creating project-specific requirement versions.
- These adaptations are not synchronised, leading to structural and semantic drift between the project templates.
- As a result, the stakeholder requirement no longer has a consistent representation across projects, introducing an inconsistency.

UNCERTAINTY 9: PERFORMANCE RANGE UNCERTAINTY



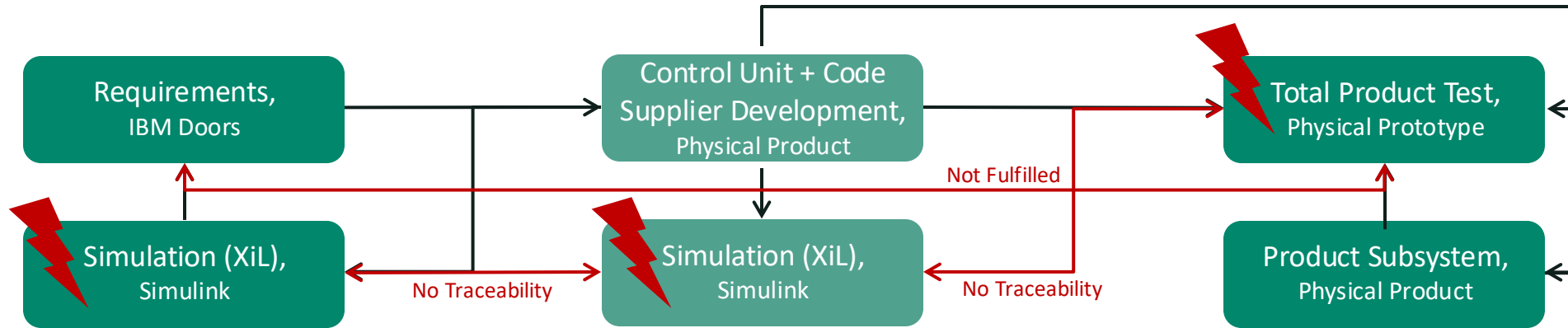
- The parameter cannot be specified precisely yet, so it is expressed as an interval.
- This range reflects variation that may arise from workload, configuration, or operational factors.
- Engineers work with this interval because a single definitive value is not available at the current development stage.

Uncertainty Representation

Location: Parameter
Source: Environment
Kind: Behaviour
Effect: continuous non-deterministic

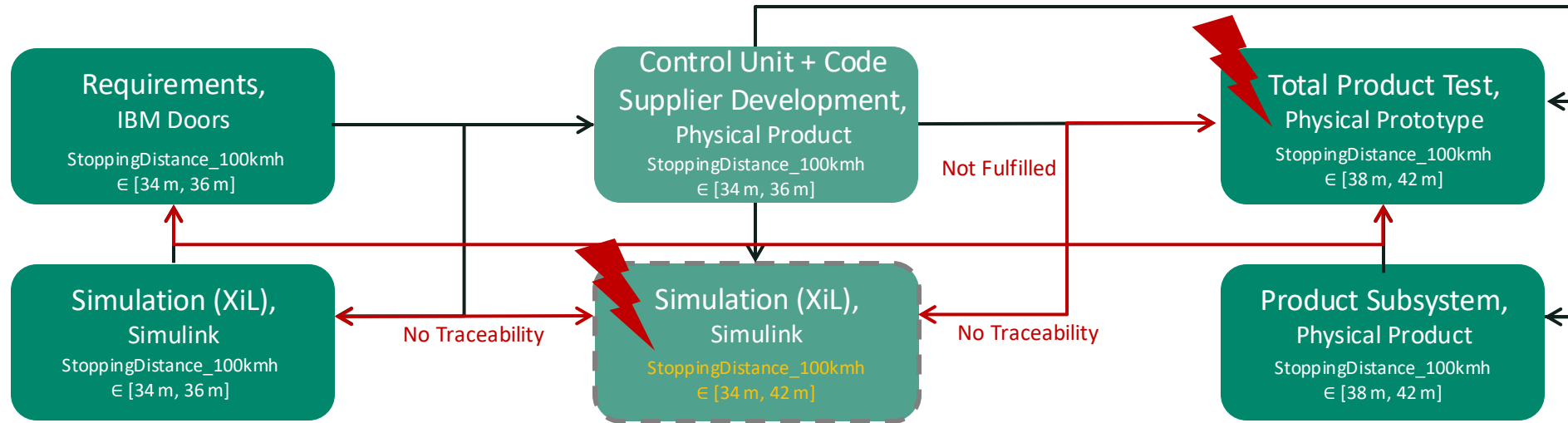
INCONSISTENCY 10:

INCONSISTENCY BETWEEN SIMULATION AND TEST BENCH



- Test results from subsystem and product tests were not transferred back into the upstream models.
- Missing traceability caused simulation and supplier development to rely on outdated evidence.
- Simulation indicated fulfilled requirements, while physical testing showed deviations.
- Broken feedback loops led to inconsistent assessments of requirement fulfillment.

UNCERTAINTY 10: CONFLICTING REGULATION BELIEF UNCERTAINTY



- Models connected by black arrows share the same interval and remain consistent.
- The left chain (Requirements, SimulationLeft, ECU) uses [34 m, 36 m].
- The hardware side (Subsystem, ProductTest) uses [38 m, 42 m].
- The right Simulink model sees both sides through broken links and must assume [34 m, 42 m].

Uncertainty Representation

Location: Outcome

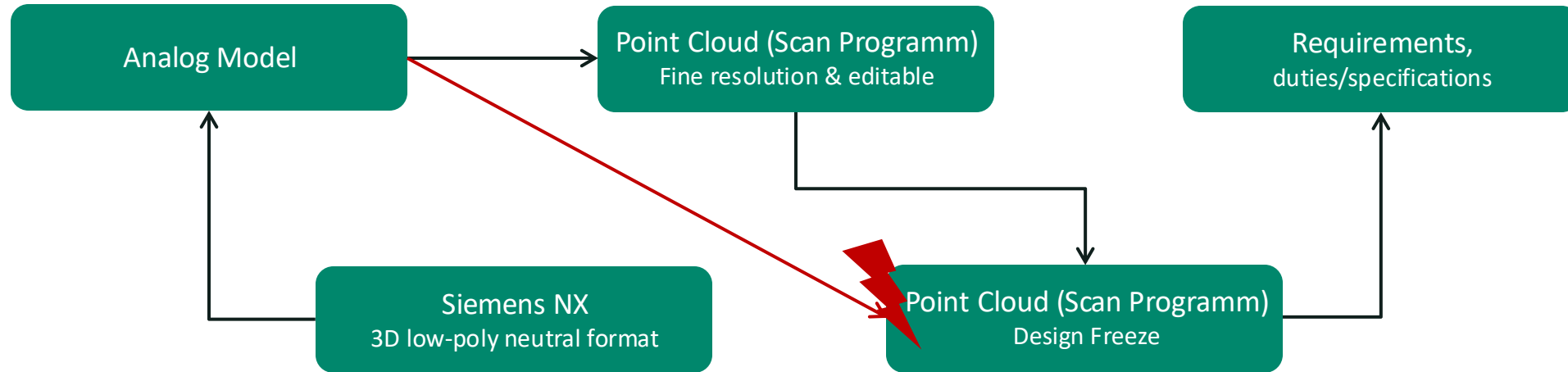
Source: Model

Kind: Behaviour

Effect: continuous non-deterministic

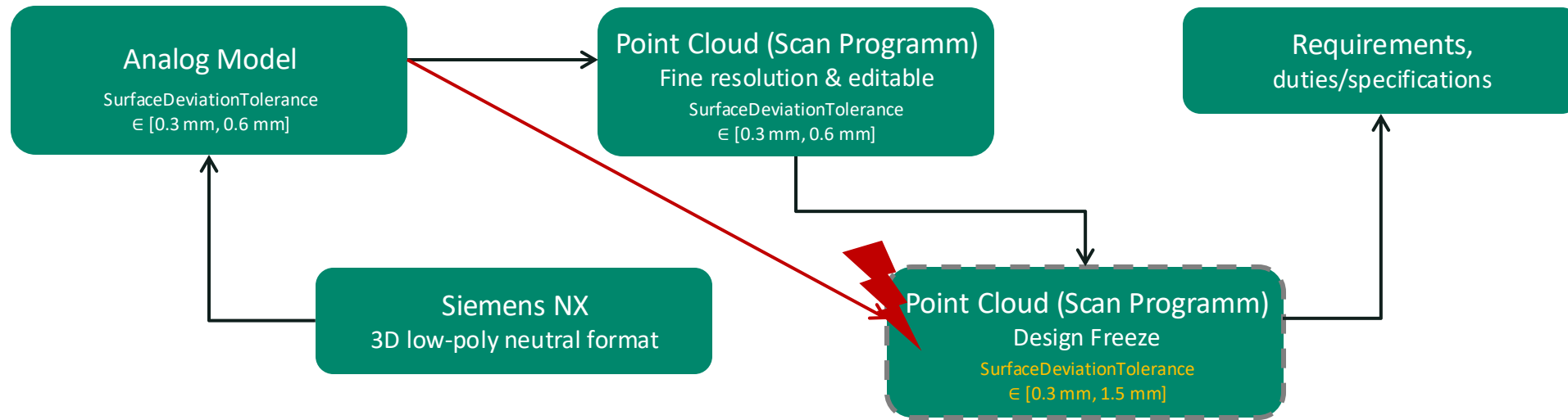
INCONSISTENCY 11:

INCONSISTENCY BETWEEN PHYSICAL SHAPE AND TECHNICAL DUTIES



- The Siemens NX model evolves based on designer updates, but the updated geometry is not re-scanned due to scanner capacity and time offset.
- The design-freeze point cloud, therefore, reflects an outdated state, while NX already contains newer modifications.
- This causes the downstream pipeline (point cloud → NX → requirements) to lose a consistent reference geometry, as artifacts are no longer synchronized.

UNCERTAINTY 11: CONFLICTING REGULATION BELIEF UNCERTAINTY



- Later design changes were not re-scanned, creating an uncertain surface deviation in the point cloud.
- The stored tolerance no longer reflects the real geometry.
- The uncertainty remains until a new scan is performed.

Uncertainty Representation

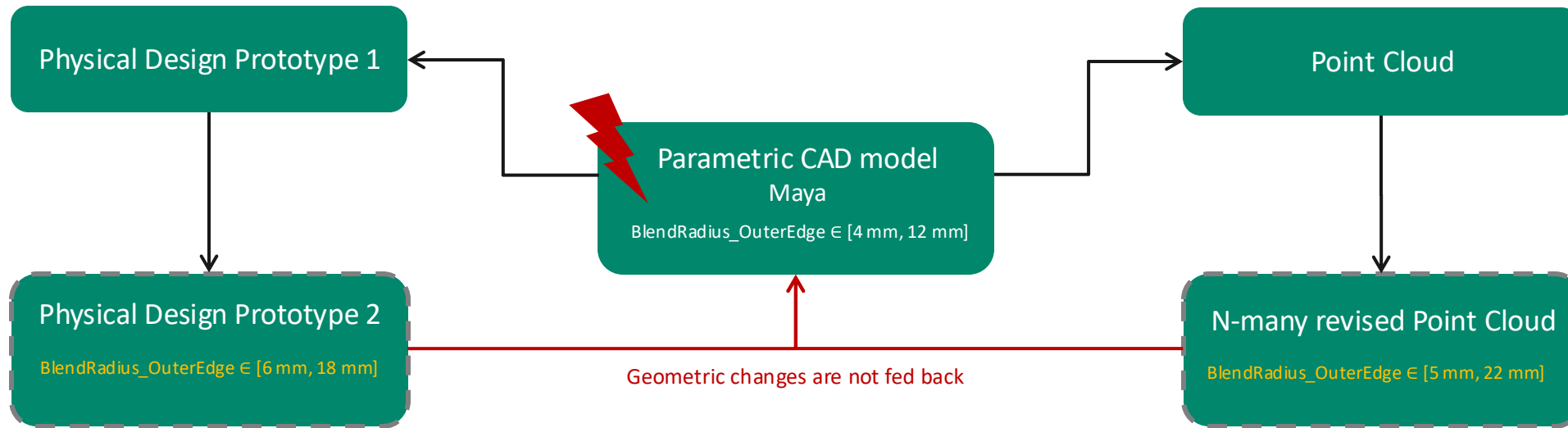
Location: Outcome

Source: Model

Kind: Behaviour

Effect: continuous non-deterministic

UNCERTAINTY 12:



- The physical prototype is manually reshaped, creating varying curvature along the outer edge.
- These geometric changes are not synchronized with the CAD model, leaving the original radius outdated.
- Multiple revised point clouds show different radius values, with no single authoritative geometry.

Uncertainty Representation

Location: Model Structure

Source: Model

Kind: Measurement

Effect: continuous non-deterministic



Uncertainty as a Source of Inconsistency

- Uncertainty exists before the models diverge.
- Teams resolve it differently.
- These differences create the inconsistency.

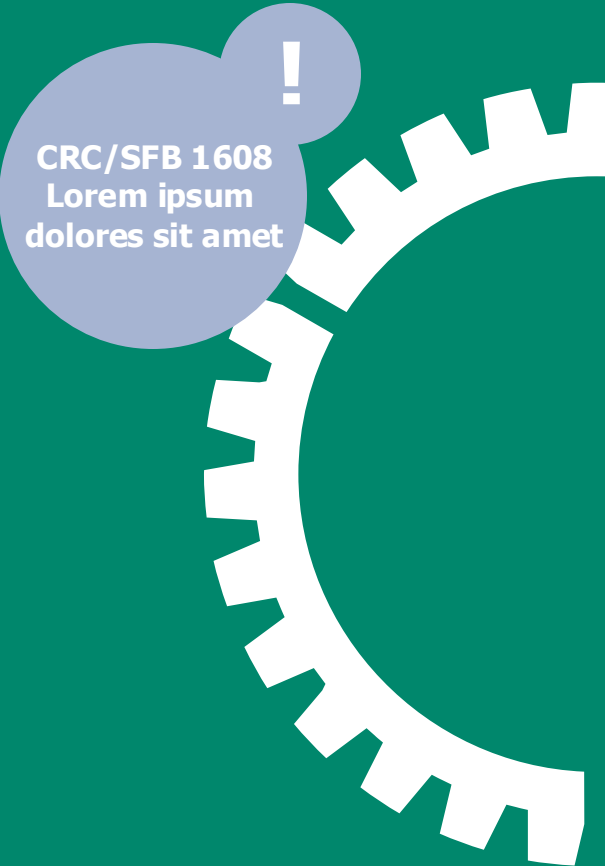
Uncertainty as Consequence of Inconsistency

- Models diverge first.
- It becomes unclear which information is correct.
- This ambiguity introduces uncertainty.

Uncertainty Independent of Inconsistency

- Some elements are inherently uncertain.
- All models represent this uncertainty consistently.
- No causal link between uncertainty and inconsistency.

VIELEN DANK!



CRC/SFB 1608
Lorem ipsum
dolores sit amet